We create chemistry

Safety data sheet

1. Identification

Product identifier

Attain TR Total Release Insecticide

Relevant identified uses of the substance or mixture and uses advised against

Relevant identified uses: crop protection product, insecticide

Details of the supplier of the safety data sheet

Company:
BASF SE
67056 Ludwigshafen
GERMANY
Operating Division Crop Protection

Telephone: +49 621 60-27777
E-mail address: Produktinformation-Pflanzenschutz@basf.com

Emergency telephone number

International emergency number:
Telephone: +49 180 2273-112

2. Hazards Identification

Classification of the substance or mixture

According to UN GHS criteria

Flam. Aerosol 1
Acute Tox. 4 (oral)
Eye Dam./Irrit. 2A
Skin Sens. 1

Carc. 2
Repr. 1B (unborn child)
STOT SE 3 (Vapours may cause drowsiness and dizziness.)
STOT RE (Nervous system) 2
Aquatic Acute 1
Aquatic Chronic 1

For the classifications not written out in full in this section the full text can be found in section 16.

Label elements

Globally Harmonized System (GHS)

Pictogram:

Signal Word:
Danger

Hazard Statement:

H222	Extremely flammable aerosol.
H229	Pressurized container: May burst if heated.
H319	Causes serious eye irritation.
H302	Harmful if swallowed.
H317	May cause an allergic skin reaction.
H336	May cause drowsiness or dizziness.
H351	Suspected of causing cancer.
H360	May damage the unborn child.
H373	May cause damage to organs (Nervous system) through prolonged or repeated exposure.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

Precautionary Statement:

P101	If medical advice is needed, have product container or label at hand.
P102	Keep out of reach of children.
P103	Read carefully and follow all instructions.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P210	Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking.
P260	Do not breathe dust/gas/mist/vapours.
P201	Obtain special instructions before use.
P271	Use only outdoors or in a well-ventilated area.
P202	Do not handle until all safety precautions have been read and understood.
P251	Do not pierce or burn, even after use.
P211	Do not spray on an open flame or other ignition source.
P272	Contaminated work clothing should not be allowed out of the workplace.
P264	Wash contaminated body parts thoroughly after handling.
P270	Do not eat, drink or smoke when using this product.

Precautionary Statements (Response):

P312	Call a POISON CENTER or physician if you feel unwell.
P305 + P351 + P338	IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308 + P313	IF exposed or concerned: Get medical attention.
P304 + P340	IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P302 + P352	IF ON SKIN: Wash with plenty of soap and water.
P333 + P313	If skin irritation or rash occurs: Get medical attention.
P330	Rinse mouth.
P391	Collect spillage.
P362 + P364	Take off contaminated clothing and wash it before reuse.
P337 + P313	If eye irritation persists: Get medical attention.

Precautionary Statements (Storage):

P403 + P233	Store in a well-ventilated place. Keep container tightly closed.
P410 + P412	Protect from sunlight. Do no expose to temperatures exceeding 50 °C/122°F.
P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

According to UN GHS criteria

Hazard determining component(s) for labelling: bifenthrin (ISO); (2-methylbiphenyl-3-yl)methyl rel-(1R,3R)-3-[(1Z)-2-chloro- 3,3,3-trifluoroprop-1-en-1-yl]- 2,2-dimethylcyclopropa necarboxylate, Propan-2-ol, 1-Methyl-2-pyrrolidone

Other hazards

According to UN GHS criteria

See section 12 - Results of PBT and vPvB assessment.

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

3. Composition/Information on Ingredients

Substances

Not applicable

Mixtures

Chemical nature

crop protection product, insecticide

Hazardous ingredients (GHS)
According to UN GHS criteria

bifenthrin (ISO); (2-methylbiphenyl-3-yl)methyl rel-(1R,3R)-3-[(1Z)-2-chloro- 3,3,3-trifluoroprop-1-en-1-yl]- 2,2-dimethylcyclopropa necarboxylate

Content (W/W): 4,08 %	Acute Tox. 3 (Inhalation - dust)
CAS Number: 82657-04-3	Acute Tox. 2 (oral)
	Skin Corr./Irrit. 2
	Eye Dam./Irrit. 2A
	Skin Sens. 1B
	Carc. 2
	STOT RE 1
	Aquatic Acute 1
	Aquatic Chronic 1
	H319, H315, H331, H300, H317, H351, H372, H400, H410

Petroleum gases, liquefied, sweetened

Content (W/W): < 50 %	Press. Gas Liquef. Gas
CAS Number: 68476-86-8	Flam. gases (incl. chem. unstable gases) 1
EC-Number: 270-705-8	H280, H220
INDEX-Number: 649-203-00-1	

Propan-2-ol

Content (W/W): < 50 %	Flam. Liq. 2
CAS Number: 67-63-0	Acute Tox. 5 (oral)
EC-Number: 200-661-7	Eye Dam./Irrit. 2A
INDEX-Number: 603-117-00-0	STOT SE 3 (drowsiness and dizziness)
	H225, H319, H303, H336

1-Methyl-2-pyrrolidone

Content (W/W): < 10 %	Flam. Liq. 4
CAS Number: 872-50-4	Acute Tox. 5 (oral)
EC-Number: 212-828-1	Skin Corr./Irrit. 2
INDEX-Number: 606-021-00-7	Eye Dam./Irrit. 2A
	Repr. 1B (unborn child)
	STOT SE 3 (irr. to respiratory syst.)
	H227, H319, H315, H303, H335, H360
	Specific concentration limit:
	STOT SE 3, irr. to respiratory syst.: >= 10 %

For the classifications not written out in full in this section the full text can be found in section 16.

4. First-Aid Measures

Description of first aid measures

First aid personnel should pay attention to their own safety. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). Immediately remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Immediately wash thoroughly with soap and water, seek medical attention.

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Most important symptoms and effects, both acute and delayed

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Indication of any immediate medical attention and special treatment needed

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Extinguishing media

Suitable extinguishing media:
foam, dry powder, carbon dioxide, water spray

Unsuitable extinguishing media for safety reasons:
water jet

Special hazards arising from the substance or mixture

Carbon monoxide, Carbon dioxide, hydrogen chloride, Hydrogen fluoride, nitrogen oxides, halogenated compounds
The substances/groups of substances mentioned can be released in case of fire. Aerosol container contains flammable gas under pressure. If product is heated above decomposition temperature, toxic vapours will be released.

Advice for fire-fighters

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Keep containers cool by spraying with water if exposed to fire. In case of fire and/or explosion do not breathe fumes. Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations.

6. Accidental Release Measures

Personal precautions, protective equipment and emergency procedures

Do not breathe vapour/spray. Use personal protective clothing. Avoid contact with the skin, eyes and clothing.

Environmental precautions

Do not discharge into the subsoil/soil. Do not discharge into drains/surface waters/groundwater.

Methods and material for containment and cleaning up

For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

7. Handling and Storage

Precautions for safe handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
Vapours may form ignitable mixture with air. Prevent electrostatic charge - sources of ignition should be kept well clear - fire extinguishers should be kept handy.

Conditions for safe storage, including any incompatibilities

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight. Keep only in the original container in a cool, dry, well-ventilated place away from ignition sources, heat or flame.

Storage stability:
Storage duration: 24 Months

Protect from temperatures above: 50 °C
The packed product must be protected against exceeding the indicated temperature.

Specific end use(s)

For the relevant identified use(s) listed in Section 1 the advice mentioned in this section 7 is to be observed.

8. Exposure Controls/Personal Protection

Control parameters

Components with occupational exposure limits

67-63-0: Propan-2-ol
872-50-4: 1-Methyl-2-pyrrolidone
68476-86-8: Petroleum gases, liquefied, sweetened

Exposure controls

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for lower concentrations or short-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

9. Physical and Chemical Properties

Information on basic physical and chemical properties

Form: aerosol

Colour:	clear
Odour:	characteristic, solvent-like
Odour threshold:	Not determined due to potential health hazard by inhalation.
pH value:	approx. 9 - 11 (10 g/l, approx. 25 °C)
Melting point:	approx. -88,5 °C Information applies to the solvent.
Boiling point:	approx. 82,3 °C Information applies to the solvent.
Flash point:	approx. -104 °C Information applies to the propellant.
Evaporation rate:	not applicable
Flammability:	not applicable
Flammability of Aerosol Products:	> 18 in (ASTM D 3065)
Lower explosion limit:	As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.
Upper explosion limit:	As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.
Ignition temperature:	399 °C Information applies to the solvent.
Vapour pressure:	approx. 60,2 hPa (25 °C) Information applies to the solvent.
Density:	approx. 0,84 g/cm3 (20 °C)
Relative vapour density (air):	not applicable
Solubility in water:	slightly soluble
Partitioning coefficient n-octanol/water (log Kow):	The statements are based on the properties of the individual components.

Information on: bifenthrin (ISO); (2-methylbiphenyl-3-yl)methyl rel-(1R,3R)-3-[(1Z)-2-chloro- 3,3,3-trifluoroprop-1-en-1-yl]- 2,2-dimethylcyclopropa necarboxylate
Partitioning coefficient n-octanol/water (log Kow): > 6

Thermal decomposition:	No decomposition if stored and handled as prescribed/indicated.
Viscosity, dynamic:	2,25 mPa.s (approx. 25 °C)
Explosion hazard:	Based on the chemical structure there is no indication of explosive properties.

Fire promoting properties: Based on its structural properties the product is not classified as oxidizing.

Other information

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

10. Stability and Reactivity

Reactivity
No hazardous reactions if stored and handled as prescribed/indicated.

Chemical stability
The product is stable if stored and handled as prescribed/indicated.

Possibility of hazardous reactions
No hazardous reactions if stored and handled as prescribed/indicated.

Conditions to avoid
See SDS section 7 - Handling and storage.

Incompatible materials

Substances to avoid:
strong acids, strong bases, strong oxidizing agents

Hazardous decomposition products

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

11. Toxicological Information

Information on toxicological effects

Acute toxicity

Assessment of acute toxicity:
Of moderate toxicity after single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
LD50 rat (oral): 1.330 mg/kg

LC50 rat (by inhalation): > 3,1 mg/l
No mortality was observed. An aerosol was tested.

LD50 rabbit (dermal): > 5.000 mg/kg

Irritation

Assessment of irritating effects:
Eye contact causes irritation. Not irritating to the skin. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant

Serious eye damage/irritation rabbit: Irritant.

Information on: Propan-2-ol
Experimental/calculated data:
Serious eye damage/irritation rabbit: Irritant. (similar to OECD guideline 405)

Respiratory/Skin sensitization

Assessment of sensitization:
Sensitization after skin contact possible. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experimental/calculated data:
modified Buehler test guinea pig: skin sensitizing

Germ cell mutagenicity

Assessment of mutagenicity:
Mutagenicity tests revealed no genotoxic potential. The product has not been tested. The statement has been derived from the properties of the individual components.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: bifenthrin (ISO); (2-methylbiphenyl-3-yl)methyl rel-(1R,3R)-3-[(1Z)-2-chloro- 3,3,3-trifluoroprop-1-en-1-yl]- 2,2-dimethylcyclopropa necarboxylate
Assessment of carcinogenicity:
Indication of possible carcinogenic effect in animal tests.

Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: 1-Methyl-2-pyrrolidone
Assessment of reproduction toxicity:
The results of animal studies gave no indication of a fertility impairing effect. As shown in animal studies, the product may cause damage to the testes after repeated high exposures that cause other toxic effects. The effects observed on testes and sperm parameters did not affect fertility in rats.

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: 1-Methyl-2-pyrrolidone
Assessment of teratogenicity:
After the uptake of small doses toxicity to development will not be expected in humans. Effects observed at maternally toxic doses.

Specific target organ toxicity (single exposure)

Assessment of STOT single:
Possible narcotic effects (drowsiness or dizziness).

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: bifenthrin (ISO); (2-methylbiphenyl-3-yl)methyl rel-(1R,3R)-3-[(1Z)-2-chloro- 3,3,3-trifluoroprop-1-en-1-yl]- 2,2-dimethylcyclopropa necarboxylate
Assessment of repeated dose toxicity:
Repeated exposure to small quantities may affect certain organs.

Information on: Propan-2-ol
Assessment of repeated dose toxicity:
The substance may cause damage to the kidney after repeated inhalation.

Information on: 1-Methyl-2-pyrrolidone
Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation. The substance may cause damage to the testes after repeated inhalation of high doses.

Aspiration hazard

No aspiration hazard expected.
The product has not been tested. The statement has been derived from the properties of the individual components.

Other relevant toxicity information

Misuse can be harmful to health.

12. Ecological Information

Toxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: bifenthrin (ISO); (2-methylbiphenyl-3-yl)methyl rel-(1R,3R)-3-[(1Z)-2-chloro- 3,3,3-trifluoroprop-1-en-1-yl]- 2,2-dimethylcyclopropa necarboxylate
Toxicity to fish:
LC50 (96 h) 0,00010 mg/l, Salmo gairdneri, syn. O. mykiss

Information on: bifenthrin (ISO); (2-methylbiphenyl-3-yl)methyl rel-(1R,3R)-3-[(1Z)-2-chloro- 3,3,3-trifluoroprop-1-en-1-yl]- 2,2-dimethylcyclopropa necarboxylate
Aquatic invertebrates:
EC50 (48 h) 0,00011 mg/l, Daphnia magna

Information on: bifenthrin (ISO); (2-methylbiphenyl-3-yl)methyl rel-(1R,3R)-3-[(1Z)-2-chloro- 3,3,3-trifluoroprop-1-en-1-yl]- 2,2-dimethylcyclopropa necarboxylate
Aquatic plants:
EC50 > 100 mg/l, algae

Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: bifenthrin (ISO); (2-methylbiphenyl-3-yl)methyl rel-(1R,3R)-3-[(1Z)-2-chloro- 3,3,3-trifluoroprop-1-en-1-yl]- 2,2-dimethylcyclopropa necarboxylate
Assessment biodegradation and elimination (H2O):
Not readily biodegradable (by OECD criteria).

Bioaccumulative potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: bifenthrin (ISO); (2-methylbiphenyl-3-yl)methyl rel-(1R,3R)-3-[(1Z)-2-chloro- 3,3,3-trifluoroprop-1-en-1-yl]- 2,2-dimethylcyclopropa necarboxylate
Assessment bioaccumulation potential:
Because of the n-octanol/water distribution coefficient (log Pow) accumulation in organisms is not to be expected.

Mobility in soil

Assessment transport between environmental compartments:
Adsorption in soil: The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: Propan-2-ol
Assessment transport between environmental compartments:
Volatility: No data available.
Adsorption in soil: Adsorption to solid soil phase is not expected.

Results of PBT and vPvB assessment

The product does not contain a substance fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

Other adverse effects

The product does not contain substances that are listed in the Montreal Protocol on substances that deplete the ozone layer.

Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

13. Disposal Considerations

Waste treatment methods

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.
Emptied gas pressure vessels must not be opened.

14. Transport Information

Land transport

ADR

UN number or ID number:	UN1950
UN proper shipping name:	AEROSOLS
Transport hazard class(es):	2.1, EHSM
Packing group:	Not applicable
Environmental hazards:	yes
Special precautions for user:	Tunnel code: D

RID

UN number or ID number:	UN1950
UN proper shipping name:	AEROSOLS
Transport hazard class(es):	2.1, EHSM
Packing group:	Not applicable
Environmental hazards:	yes
Special precautions for user:	None known

Inland waterway transport

ADN

UN number or ID number:	UN1950
UN proper shipping name:	AEROSOLS
Transport hazard class(es):	2.1, EHSM
Packing group:	Not applicable
Environmental hazards:	yes
Special precautions for user:	None known

Transport in inland waterway vessel
Not evaluated

Sea transport

IMDG

UN number or ID number:	UN 1950
UN proper shipping name:	AEROSOLS
Transport hazard class(es):	2.1, EHSM
Packing group:	Not applicable
Environmental hazards:	yes Marine pollutant: YES
Special precautions for user:	EmS: F-D; S-U

Air transport

IATA/ICAO

UN number or ID number:	UN 1950
UN proper shipping name:	AEROSOLS, FLAMMABLE
Transport hazard class(es):	2.1
Packing group:	Not applicable
Environmental hazards:	No Mark as dangerous for the environment is needed
Special precautions for	None known

user:

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information

Safety, health and environmental regulations/legislation specific for the substance or mixture

To avoid risks to man and the environment, comply with the instructions for use.

16. Other Information

Full text of classifications, hazard symbols and hazard statements, if mentioned in section 2 or 3:

Flam. Aerosol	Flammable aerosols
Acute Tox.	Acute toxicity
Eye Dam./Irrit.	Serious eye damage/eye irritation
Skin Sens.	Skin sensitization
Carc.	Carcinogenicity
Repr.	Reproductive toxicity
STOT SE	Specific target organ toxicity — single exposure
STOT RE	Specific target organ toxicity — repeated exposure
Aquatic Acute	Hazardous to the aquatic environment - acute
Aquatic Chronic	Hazardous to the aquatic environment - chronic
Skin Corr./Irrit.	Skin corrosion/irritation
Press. Gas	Gases under pressure
Flam. gases (incl. chem. unstable gases)	Flammable gases (including chemically unstable gases)
Flam. Liq.	Flammable liquids
H319	Causes serious eye irritation.
H315	Causes skin irritation.
H331	Toxic if inhaled.
H300	Fatal if swallowed.
H317	May cause an allergic skin reaction.
H351	Suspected of causing cancer.
H372	Causes damage to organs through prolonged or repeated exposure.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.
H280	Contains gas under pressure; may explode if heated.
H220	Extremely flammable gas.
H225	Highly flammable liquid and vapour.
H303	May be harmful if swallowed.
H336	May cause drowsiness or dizziness.
H227	Combustible liquid.
H335	May cause respiratory irritation.
H360	May damage the unborn child.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification

agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.

Vertical lines in the left hand margin indicate an amendment from the previous version.